1. ค่าคลาดเคลื่อนสัมบูรณ์ (E_{abs}) :

$$E_{abs} = |\text{ ค่าจริง } - \text{ ค่าประมาณ }|$$

2. ค่าคลาดเคลื่อนสัมพัทธ์ (E_{rel}):

$$E_{rel} = \frac{|\text{ ค่าจริง } - \text{ ค่าประมาณ }|}{|\text{ ค่าจริง }|}$$

3. ร้อยละของค่าความคลาดเคลื่อนสัมพัทธ์ (ε_t) :

$$\varepsilon_t = \frac{|\text{ ค่าจริง } - \text{ ค่าประมาณ }|}{|\text{ ค่าจริง }|} \times 100\%$$

4. Linear interpolation:

$$f_1(x) = f(x_0) + \frac{f(x_1) - f(x_0)}{x_1 - x_0}(x - x_0)$$

5. Quadratic Interpolation:
   $f_2(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1)$
   เมื่อ

   (a) $b_0 = f(x_0)$

   (b) $b_1 = \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}$

   (c)

$$b_2 = \frac{\dfrac{f(x_2) - f(x_1)}{x_2 - x_1} - \dfrac{f(x_1) - f(x_0)}{x_1 - x_0}}{x_2 - x_0}$$

6. General Form of Newton's Interpolating Polynomials:
   รูปทั่วไปของสมการพหุนามอันดับที่ n ผ่านจุดข้อมูล $n + 1$ จุด คือ

$$f_n(x) = b_0 + b_1(x - x_0) + b_2(x - x_0)(x - x_1) + \cdots + b_n(x - x_0)(x - x_1) \cdots (x - x_{n-1}) \tag{1}$$

   โดยที่

$$\begin{aligned} b_0 &= f(x_0) \\ b_1 &= f[x_1, x_0] \\ b_2 &= f[x_2, x_1, x_0] \\ &\vdots \\ b_n &= f[x_n, x_{n-1}, x_{n-2}, ..., x_1, x_0] \end{aligned}$$

   เมื่อ $f[x_n, x_{n-1}, ..., x_1, x_0]$ คือ ผลต่างจากการแบ่งย่อยอันดับที่ n (nth finite divided difference) นิยามโดย

$$f[x_n, x_{n-1}, ..., x_1, x_0] = \frac{f[x_n, x_{n-1}, ..., x_2, x_1] - f[x_{n-1}, x_{n-2}, ..., x_1, x_0]}{x_n - x_0}$$

7. Lagrange Interpolating Polynomials:

$$f_n(x) = \sum_{i=0}^{n} L_i(x) f(x_i)$$

เมื่อ

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^{n} \frac{x - x_j}{x_i - x_j}$$

8. วิธีสี่เหลี่ยมคางหมู (Trapezoidal Rule)

$$I = (b-a)\frac{f(b) + f(a)}{2}$$

9. วิธีสี่เหลี่ยมคางหมูหลายรูป (Multiple Application Trapezoidal Rule)

$$I = (b-a)\frac{\left[f(x_0) + 2\sum_{i=1}^{n-1} f(x_i) + f(x_n)\right]}{2n}$$

10. วิธีซิมสัน 1/3 (Simpson's 1/3 Rule)

$$I = (b-a)\frac{f(x_0) + 4f(x_1) + f(x_2)}{6}$$

เมื่อ $h = (b-a)/2$

11. วิธีซิมสัน 1/3 หลายรูป (Multiple Application Simpson's 1/3 Rule)

$$I = (b-a)\frac{f(x_0) + 4\sum_{i=1,3,5}^{n-1} f(x_i) + 2\sum_{i=2,4,6}^{n-2} f(x_i) + f(x_n)}{3n}$$

12. วิธีซิมสัน 3/8 (Simpson's 3/8 Rule)

$$I = \frac{b-a}{8}\left[f(x_0) + 3f(x_1) + 3f(x_2) + f(x_3)\right]$$

เมื่อ $h = (b-a)/3$

13. Forward finite-divided-difference formulas

First Derivative	Error
$f'(x_i) = \dfrac{f(x_{i+1}) - f(x_i)}{h}$	$O(h)$
$f'(x_i) = \dfrac{-f(x_{i+2}) + 4f(x_{i+1}) - 3f(x_i)}{2h}$	$O(h^2)$
Second Derivative	
$f''(x_i) = \dfrac{f(x_{i+2}) - 2f(x_{i+1}) + f(x_i)}{h^2}$	$O(h)$
$f''(x_i) = \dfrac{-f(x_{i+3}) + 4f(x_{i+2}) - 5f(x_{i+1}) + 2f(x_i)}{h^2}$	$O(h^2)$
Third Derivative	
$f'''(x_i) = \dfrac{f(x_{i+3}) - 3f(x_{i+2}) + 3f(x_{i+1}) - f(x_i)}{h^3}$	$O(h)$
$f'''(x_i) = \dfrac{-3f(x_{i+4}) + 14f(x_{i+3}) - 24f(x_{i+2}) + 18f(x_{i+1}) - 5f(x_i)}{2h^3}$	$O(h^2)$
Fourth Derivative	
$f''''(x_i) = \dfrac{f(x_{i+4}) - 4f(x_{i+3}) + 6f(x_{i+2}) - 4f(x_{i+1}) + f(x_i)}{h^4}$	$O(h)$
$f''''(x_i) = \dfrac{-2f(x_{i+5}) + 11f(x_{i+4}) - 24f(x_{i+3}) + 26f(x_{i+2}) - 14f(x_{i+1}) + 3f(x_i)}{h^4}$	$O(h^2)$

รูปภาพ 1: Forward finite-divided-difference formulas

14. Backward finite-divided- difference formulas

First Derivative — Error

$$f'(x_i) = \frac{f(x_i) - f(x_{i-1})}{h} \qquad O(h)$$

$$f'(x_i) = \frac{3f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{2h} \qquad O(h^2)$$

Second Derivative

$$f''(x_i) = \frac{f(x_i) - 2f(x_{i-1}) + f(x_{i-2})}{h^2} \qquad O(h)$$

$$f''(x_i) = \frac{2f(x_i) - 5f(x_{i-1}) + 4f(x_{i-2}) - f(x_{i-3})}{h^2} \qquad O(h^2)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_i) - 3f(x_{i-1}) + 3f(x_{i-2}) - f(x_{i-3})}{h^3} \qquad O(h)$$

$$f'''(x_i) = \frac{5f(x_i) - 18f(x_{i-1}) + 24f(x_{i-2}) - 14f(x_{i-3}) + 3f(x_{i-4})}{2h^3} \qquad O(h^2)$$

Fourth Derivative

$$f''''(x_i) = \frac{f(x_i) - 4f(x_{i-1}) + 6f(x_{i-2}) - 4f(x_{i-3}) + f(x_{i-4})}{h^4} \qquad O(h)$$

$$f''''(x_i) = \frac{3f(x_i) - 14f(x_{i-1}) + 26f(x_{i-2}) - 24f(x_{i-3}) + 11f(x_{i-4}) - 2f(x_{i-5})}{h^4} \qquad O(h^2)$$

รูปภาพ 2: Backward finite-divided- difference formulas

15. Centered finite-divided- difference formulas

First Derivative | Error

$$f'(x_i) = \frac{f(x_{i+1}) - f(x_{i-1})}{2h} \qquad O(h^2)$$

$$f'(x_i) = \frac{-f(x_{i+2}) + 8f(x_{i+1}) - 8f(x_{i-1}) + f(x_{i-2})}{12h} \qquad O(h^4)$$

Second Derivative

$$f''(x_i) = \frac{f(x_{i+1}) - 2f(x_i) + f(x_{i-1})}{h^2} \qquad O(h^2)$$

$$f''(x_i) = \frac{-f(x_{i+2}) + 16f(x_{i+1}) - 30f(x_i) + 16f(x_{i-1}) - f(x_{i-2})}{12h^2} \qquad O(h^4)$$

Third Derivative

$$f'''(x_i) = \frac{f(x_{i+2}) - 2f(x_{i+1}) + 2f(x_{i-1}) - f(x_{i-2})}{2h^3} \qquad O(h^2)$$

$$f'''(x_i) = \frac{-f(x_{i+3}) + 8f(x_{i+2}) - 13f(x_{i+1}) + 13f(x_{i-1}) - 8f(x_{i-2}) + f(x_{i-3})}{8h^3} \qquad O(h^4)$$

Fourth Derivative

$$f''''(x_i) = \frac{f(x_{i+2}) - 4f(x_{i+1}) + 6f(x_i) - 4f(x_{i-1}) + f(x_{i-2})}{h^4} \qquad O(h^2)$$

$$f''''(x_i) = \frac{-f(x_{i+3}) + 12f(x_{i+2}) - 39f(x_{i+1}) + 56f(x_i) - 39f(x_{i-1}) + 12f(x_{i-2}) - f(x_{i-3})}{6h^4} \qquad O(h^4)$$

รูปภาพ 3: Centered finite-divided- difference formulas